Artur Tigirlas

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Education

Bachelor of Technology (Co-op) – Automotive and Vehicle Engineering Technology Hamilton, ON
McMaster University *2024 – 2029*

- Currently enrolled in Level II of McMaster University's Automotive Engineering Technology program.
- Working toward a **Bachelor of Technology Degree**, **Advanced Diploma in Mechanical Engineering Technology**, and an **ACBSP & CIM accredited Business Management Certificate**.
- Coursework includes AutoCAD, CAD, Machine Elements, Materials Engineering, Thermodynamics, Heat Transfer, Electronics, and Engineering Economics.

Experience

Operations Crew Lead Apr 2025 – Jun 2025
Willowbrook Nurseries *Fenwick, ON*

- Led daily crew operations while maintaining organized workflow and meeting production targets.
- Verified inventory records, shipment documentation, and material quantities for daily operations.
- Tracked workflow progress and maintained detailed documentation supporting planning and coordination.

Office Assistant Student – Automation, Accounting & Operations Support May 2026 – Present
Niagara Ina Grafton Gage Village *St. Catharines, ON*

- Developed a Python OCR desktop application using Tkinter, PyMuPDF, Pillow, pytesseract, threading, JSON, and openpyxl to digitize scanned invoice PDFs, validate extracted data, and export formatted Excel reports.
- Built reusable supplier-specific OCR templates with full-column capture, automatic row creation, fill-down tools, duplicate invoice warnings, undo history, and save/reopen functionality to streamline invoice processing.
- Engineered a macro-enabled Excel purchasing and budget-management system using VBA, Power Query, formulas, data validation, charts, and interactive worksheet controls.
- Built VBA routines to import large purchasing datasets from external Excel files, validate headers and field types, detect duplicate transactions, append records, and summarize import errors.
- Created dynamic search, reporting, and budget dashboard tools to analyze spending by company, category, item keyword, year, and date range across multiple worksheets.
- Automated repetitive Accounting, HR, and Kitchen/Nutrition workflows using Excel macros, formulas, Power Query transformations, and query-driven spreadsheets.
- Updated historical budget records by reviewing prior-year account files, invoices, account numbers, expense categories, utility billings, and facility spending data across multiple buildings.
- Calculated hydro and water spending across multiple buildings and organized total facility billings to support budget tracking and financial review.
- Updated and verified care plan documentation for approximately **400 residents**, cross-referencing medical records and information sheets to identify discrepancies and ensure accurate resident records.
- Supported Accounting, HR, Kitchen/Nutrition, Long-Term Care, and Assisted Living departments by maintaining records, updating documents, archiving files, and completing cross-functional projects.
- Updated employee handbooks and internal manuals by reviewing outdated information, revising content, and preparing documents for staff use.
- Supported a PhD Director of Care-led manuscript by synthesizing academic literature on burnout, sleep quality, psychological distress, resilience, workload, staffing shortages, shift work, and workplace stress among nurses, PSWs, and older-adult care staff.
- Created resident service materials and researched government funding opportunities for resident activity-space improvements, including program eligibility and potential funding sources.

Brand Ambassador Oct 2025 – Present
American Eagle *St. Catharines, ON*

- Assisted customers by identifying needs and recommending products based on fit, style preferences, and purchasing decisions.
- Maintained organized displays, stock areas, and store presentation supporting daily operations.
- Communicated product information clearly to support customer decisions and service quality.

- Maintained store readiness through cleaning, restocking, and routine operational tasks.

Sales Associate Jun 2023 – Apr 2025

Peavey Mart *St. Catharines, ON*

- Assisted customers with hardware and equipment selection using technical product requirements.
- Maintained organized inventory systems, product records, and department organization.
- Performed equipment troubleshooting and basic mechanical assessment.
- Developed practical knowledge of industrial tools, mechanical systems, and material applications.
- Communicated technical information clearly with customers and team members.

PROJECTS

Inline-5 Volvo Engine Model | CATIA V5 Feb 2026 – Apr 2026

- Collaborated in a team to reverse-model and assemble a complete inline-5 Volvo engine in **CATIA V5** using reference models, part drawings, and assembly constraints.
- Modeled and refined major engine components including the engine block, engine base, crankshaft, flywheel, plain bearing crank, intake camshaft, and exhaust camshaft.
- Set up initial kinematics for the valve system and piston orientation, correcting alignment issues to improve motion accuracy and assembly fitment.
- Verified part fitment, assembly accuracy, and component alignment while assisting teammates with part modeling and deadline-critical revisions.
- Helped coordinate team meetings, track part completion, and support final integration, contributing to a final project grade of **99.5%**.

Radiator Heat Transfer Test Rig | SolidWorks, Thermal Analysis 2025

- Designed and tested a radiator cooling rig to evaluate the effects of fan direction and pump voltage on cooling rate, flow rate, heat transfer, and system efficiency.
- Designed fan mounts in **SolidWorks** and used calipers to verify dimensions, fitment, and alignment during physical assembly.
- Collected temperature and flow-rate data at 8V and 12V pump settings, comparing fan-inward and fan-outward configurations over timed test runs.
- Calculated heat transfer using mass flow rate, specific heat capacity, and temperature change to compare cooling performance across test conditions.
- Analyzed results showing 12V produced an approximately **8% higher flow rate** than 8V, while 8V showed a **20–25% drop in heat transfer**.

Oil Pump Rig Design | SolidWorks Jan 2026 – Feb 2026

- Collaborated in a five-person team to design a mechanical test rig using project drawings and technical specifications.
- Created detailed **3D CAD models**, assemblies, and manufacturing drawings using **SolidWorks**.
- Performed engineering calculations, dimensional checks, and drawing revisions supporting design accuracy.
- Prepared technical documentation and maintained design notes for project reviews and revisions.

Electricity Billing System Simulation | C++ Jan 2025 – Apr 2025

- Developed a multi-user system in **C++** supporting structured workflow execution.
- Applied object-oriented programming, testing, and debugging to validate system performance.
- Collaborated within a five-person team to coordinate development milestones.
- Led testing and final integration resulting in successful system operation.

Automotive Restoration Project 2024 – 2025

- Performed mechanical repair, diagnostics, and component installation in a workshop environment.
- Used precision measurement tools for adjustments and dimensional verification.
- Followed safety procedures and organized work practices during repairs.

Human Resources Recruitment Project Jan 2026 – Mar 2026

- Worked in a four-person team to evaluate recruitment methods, applicant communication, and selection steps.
- Reviewed job postings, applicant screening, and interview planning as part of the hiring process.
- Built a stronger understanding of the detail, organization, and decision-making required in recruitment.

Management Interview Project Sep 2025 – Dec 2025

- Worked in a four-person team to prepare interview questions and conduct a management interview.

- Collected and analyzed responses by comparing workplace practices with management theories.
- Strengthened communication, data gathering, and analytical skills through structured review.

TECHNICAL SKILLS

CAD: AutoCAD, SolidWorks, CATIA, 3D Modeling, Assemblies, Engineering Drawings
Programming: Python, C++, Java, VBA, Object-Oriented Programming, Automation Scripting
Data & Automation: Excel Macros, Power Query, Formulas, Data Validation, CSV/Excel Export, JSON, OCR, Tesseract, openpyxl
Python Libraries: Tkinter/ttk, PyMuPDF, Pillow, pytesseract, threading, queue
Engineering: Mechanical Design, Engineering Calculations, Technical Documentation, Drawing Interpretation, GD&T (basic)
Software: Microsoft Excel, Word, PowerPoint, Outlook
Mechanical & Tools: Automotive Systems, Mechanical Assembly, Precision Measurement Tools, Hand & Power Tools, Troubleshooting, Safety Compliance